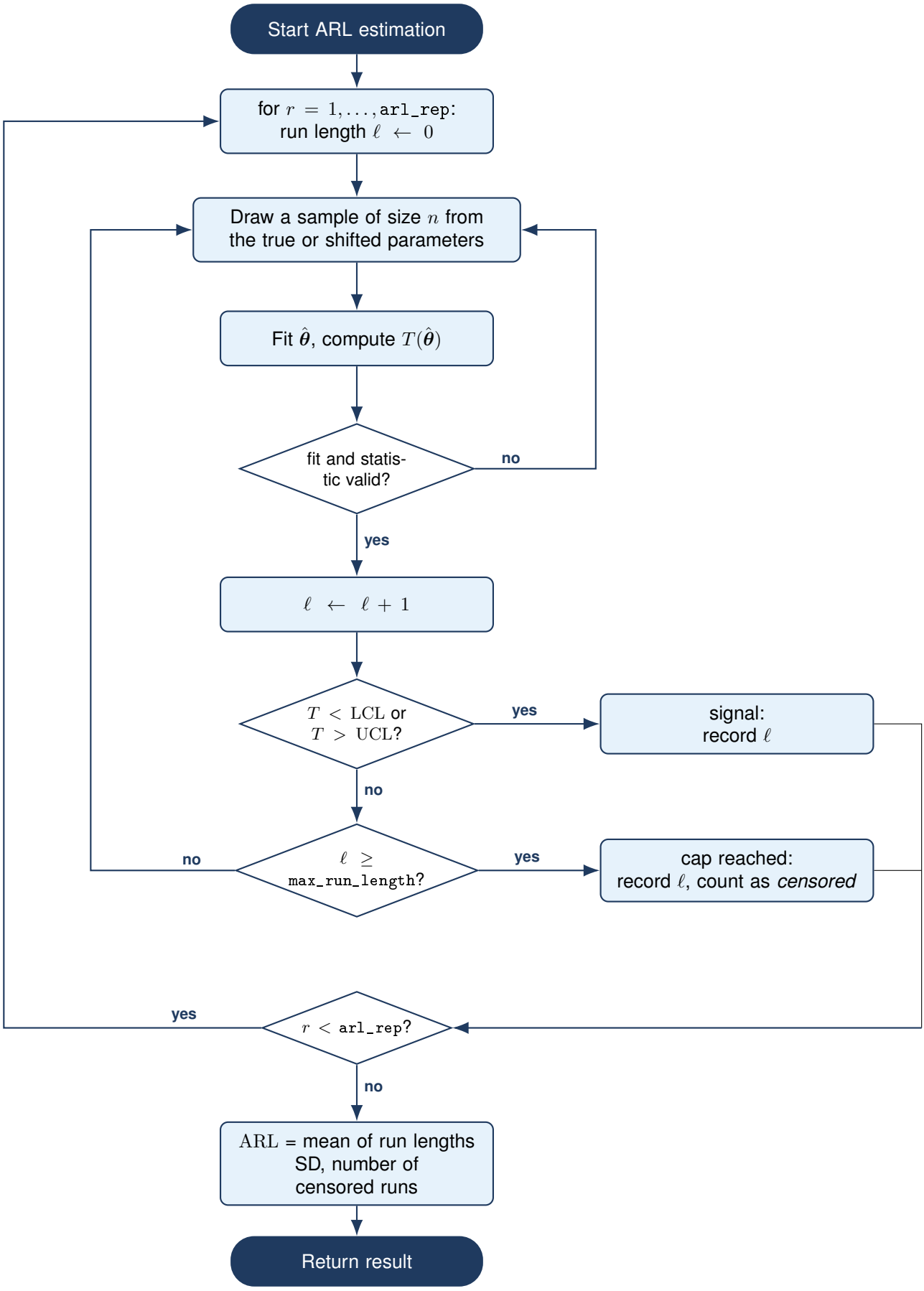

Start ARL estimation
for $r = 1, \ldots,$ arl_rep:
run length $\ell \leftarrow 0$
Draw a sample of size n from
the true or shifted parameters
Fit $\hat{\boldsymbol{\theta}}$, compute $T(\hat{\boldsymbol{\theta}})$
fit and statis-
tic valid?
no
yes
$\ell \leftarrow \ell + 1$
$T < \mathrm{LCL}$ or
$T > \mathrm{UCL}$?
yes
signal:
record ℓ
no
$\ell \geq$
max_run_length?
no
yes
cap reached:
record ℓ, count as *censored*
$r <$ arl_rep?
yes
no
ARL = mean of run lengths
SD, number of
censored runs
Return result